Parabolic SAR

Indicators & Analysis / Trend | beginner | 4 min read | DeepCharts Help Center | August 24, 2026

The Parabolic SAR is a technical analysis indicator developed by J. Welles Wilder that helps identify trend reversals and establish entry and exit points for trading. It appears as a series of points positioned above or below price, depending on the direction of the trend: below prices during uptrends, above them during downtrends.

SAR stands for "stop and reverse" — the dots trail price at an accelerating pace, and when price touches them the indicator flips to the opposite side, marking a potential reversal.

What it is

Parabolic SAR answers the question: where is my trailing stop, and has the trend reversed? This indicator is similar to the moving average, the only difference being that the Parabolic SAR moves at a more accelerated pace and can change its position relative to price. The longer a trend runs, the faster the dots close in on price — which is what gives the trail its parabolic shape and forces an eventual exit.

[SCREENSHOT PLACEHOLDER] A candlestick chart with Parabolic SAR dots below price during an uptrend, flipping above price after a reversal; the flip point clearly visible `dc-en-parabolic-sar-01.png`

When to use it

- Trailing a stop behind a trending position — the dots give a concrete, mechanically updated stop level each bar.
- Timing exits: when price touches the dots, the trend that carried the position is considered over.
- Identifying reversal points in markets that trend cleanly.
- Confirming direction alongside slower tools — dots below price agree with a bullish read, dots above with a bearish one.

Quick start

1. Open a chart — see Open Your First Chart.
2. Click the bar-chart icon in the chart's top-left icon row to open the **Indicators** panel, then click **Indicators** to open the full **Indicator List**.
3. Search for **Parabolic SAR** and click **+** to add it.
4. Click the gear icon to open its settings.
5. Starter configuration: keep **Step** at the default 0.02 and **Maximum** at the default 0.20 — Wilder's original values, and the standard starting point on any timeframe.

[SCREENSHOT PLACEHOLDER] Parabolic SAR settings dialog showing the Parameters section with Step = 0.02 and Maximum = 0.20, and the Subgraph styling section below `dc-en-parabolic-sar-02.png`

How to read it

- **Dots below price** — uptrend. Each new bar the dots rise, trailing the move.
- **Dots above price** — downtrend, with the dots descending toward price.
- **A flip** (dots jump to the other side) signals a stop-and-reverse: the prior trend is considered finished. Aggressive traders reverse position on the flip; conservative traders treat it as an exit only.
- **Dot spacing** shows acceleration: dots pulling away from each other mean the acceleration factor is climbing because the trend keeps making new extremes.

• In sideways markets the dots flip constantly — the indicator is explicitly designed for trending conditions and produces its worst signals in ranges.

Settings reference

Parameters

Setting	What it does
Step	The acceleration factor that determines how quickly the SAR catches up to price. Default: 0.02. Raising it makes the trail tighten faster (earlier exits, more whipsaws); lowering it gives the trend more room.
Maximum	The maximum acceleration factor value that can be reached. Default: 0.20. Caps how aggressively the dots can converge on price in extended trends.

Subgraph

Setting	What it does
Parabolic SAR Color	Color of the plotted dots.
Secondary Color	Optional secondary color.
Subgraph Style	Display style of the series.
Line Style	Line appearance definition.
Line Width	Size/thickness of the plot.
Use Secondary Axis	Displays the indicator on an alternative axis.

Tips and common mistakes

• **Do not trade every flip.** In a range the SAR flips repeatedly and each flip fails; use it only when a trend is established, or filter with Average Directional Index (ADX) to confirm trend strength first.

• **Tune Step to the market's character.** The 0.02 / 0.20 defaults are Wilder's originals; a smaller **Step** suits volatile instruments that need breathing room, a larger one suits fast momentum moves you want to protect quickly.

• **Treat the dot as a stop level, not a prediction.** The SAR tells you where the trend is invalidated — it says nothing about where price goes next.

• Compare it with Super Trend: both trail the trend, but Super Trend's ATR basis adapts to volatility while SAR's acceleration adapts to trend duration.

Related articles

• Super Trend

• Moving Average

• Average Directional Index (ADX)

• Average True Range (ATR)

• Different Types of Input Data for Indicators

• How to Change the Layout of Indicators